Ajay Venkatesh

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Education

New York University, New York, NY *Aug 2024 – May 2026*
Master of Science in Computer Engineering (GPA: 3.9/4.0)
Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- **CampusMesh** *Jan 2026 – Present*
  AI Software Engineer New York, US
  - Shipped production **LLM API integrations** and **RAG pipelines** for CampusIQ (AI Assistant), using **embedding-based semantic search** to improve contextual accuracy by **30%** for clinician-adjacent academic workflows.
  - Built scalable **Python** and **TypeScript/Node.js** backend services powering real-time workflows across messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
  - Deployed **Docker**-ready **serverless services** on AWS (**Lambda, API Gateway, S3, CloudFront**) across **4 environments** with **KMS/SSM** secrets and **least-privilege IAM**.
- **Amazon** *May 2025 – Aug 2025*
  Software Development Engineer Intern Seattle, US
  - Integrated production **LLM-powered RAG** with an **agentic workflow** that retrieves operational logs, reasons over incidents, and generates context-aware remediation steps for distributed backend systems.
  - Built a high-throughput triage platform on **AWS ECS Fargate, SQS, SNS** with **AWS CDK** IaC, reducing response latency by **50%** and supporting reliable daily release cadences.
  - Implemented a **Model Context Protocol (MCP)** server for tool-augmented **LLM API** retrieval, improving query efficiency and compute utilization under peak load.
- **New York University (Novel AI Technologies, Inc.)** *Aug 2024 – Present*
  AI Software Engineer New York, US
  - Architected an **NSF-funded ($100K)** AI-powered **healthcare platform** for **iOS** and **Android**, ingesting real-time wearable telemetry and deploying cloud-hosted models for patient-facing clinical insights.
  - Built full-stack **AI-powered features** (**React, TypeScript, Python, Node.js, REST APIs**) with **GPT-4 + LangChain** document workflows, **RBAC**, and row-level security for regulated healthcare clients.
  - Engineered **Python ETL/data pipelines** into **PostgreSQL**; processed **250K+** records and powered **AWS QuickSight** reporting infrastructure for operational monitoring.
- **PricewaterhouseCoopers (PwC)** *Aug 2021 – Aug 2024*
  Senior Software Engineer Bengaluru, India
  - Designed a distributed backend on **Microsoft Azure** for a **health platform**, supporting **ML inference pipelines** and integrations with **Microsoft Dynamics**.
  - Built an **NLP-powered OCR pipeline** using **LayoutLM** and Azure Form Recognizer to automate clinical document extraction, reducing processing time by **75%**.
  - Collaborated with product and operations stakeholders in agile delivery; mentored engineers on design and testing, reducing production bugs by **25%**.

Projects

- **Hospital Management System (Healthcare Full-Stack)** | *Python, React, TypeScript, Node.js, MongoDB, AWS SageMaker*
  - Built a **full-stack healthcare platform** managing appointments, patient records, and lab reports with **role-based access control** and API-driven clinical workflows.
  - Fine-tuned a **Hugging Face transformer model** reaching **88% accuracy** and deployed inference on **AWS SageMaker** for real-time diagnostic support.
- **InviLite (LLM-Powered Backend Platform)** | *Python, AWS Bedrock, FAISS, Kinesis, Lambda, Docker, Kubernetes*
  - Built a production **RAG system** (SVM + FAISS + **Bedrock LLM APIs**), cutting incident resolution from **1–2 hours to minutes** and lifting classification accuracy from **56% to 92%**.
  - Architected cloud-native **Python** ingestion pipelines (Kinesis, Lambda, S3, Glue, DynamoDB) with monitoring hooks for scalable backend operations.

Skills

- **Programming Languages:** Python, TypeScript, JavaScript (ES6), Java, SQL, C++
- **AI/ML & LLM APIs:** GPT-4, AWS Bedrock, LangChain, RAG (FAISS), Hugging Face, PyTorch, LLM Fine-tuning, NLP, Embeddings
- **Web & Backend:** React, Node.js, Express.js, REST API, Microservices, Full-Stack Development
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB
- **Cloud & DevOps:** Docker, Kubernetes, AWS (Lambda, SageMaker, Bedrock, ECS, S3, CDK, IAM), Azure, CI/CD
- **Tools & Platforms:** Git, Linux, Jupyter, Postman, Model Context Protocol (MCP)

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*
An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals